AI-Driven Food Waste Reduction in Supermarkets

Using Time-Series Demand Forecasting

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Societal Challenge

Problem:

- Supermarket food waste from demand mismatch
- Perishable items overstocked to avoid empty shelves
- Unsold products expire and are discarded
- Causes environmental, economic, and social harm

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Why It Matters (Evidence)

- Around one-third of food is lost/wasted globally (FAO)
- Food waste leads to high emissions and wasted resources (UNEP)
- Retail demand planning can reduce waste
- Reducing waste improves sustainability and saves costs

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Aim, Research Question, LLM Use

Aim:

- Develop a forecasting-based decision tool to reduce perishable food waste while maintaining availability.

Research Question:

- How can time-series forecasting reduce supermarket food waste without increasing stockouts?

LLM Use:

- Brainstorming initial possibilities
- Kept only credible, source-checked ideas

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Data Inputs and Features

Planned Data Inputs

- Sales history
- Promotions
- Calendar effects
- Weather data
- Shelf-life

Feature Examples

- Lag demand
- Rolling averages
- Promo-season interaction

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Proposed Method

1. Baseline forecast (naive / moving average)
2. Improved model (Prophet or XGBoost)
3. Generate order recommendations

Output:
Suggested daily order quantities

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Evaluation, Expected Impact, Risks

Evaluation Metrics

- Waste reduction
- Stockout
- Cost savings estimate
- MAE, RMSE, MAPE

Expected Impact

- Lower food waste
- Better inventory management
- Lower environmental footprint

Risks

- Data quality issues
- Sudden demand shocks
- Under-forecasting (buffer + human oversight)

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Conclusion & Feedback Request

Conclusion

- Practical AI/data science approach for food waste
- Suitable for Task 2.1, expandable for Task 2.2

Feedback requested

- Is the scope appropriate?
- Prophet vs XGBoost for this level?
- Recommended public datasets for supermarket perishables?

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References

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- Goodfellow, Bengio, & Courville (2016). *Deep Learning*.

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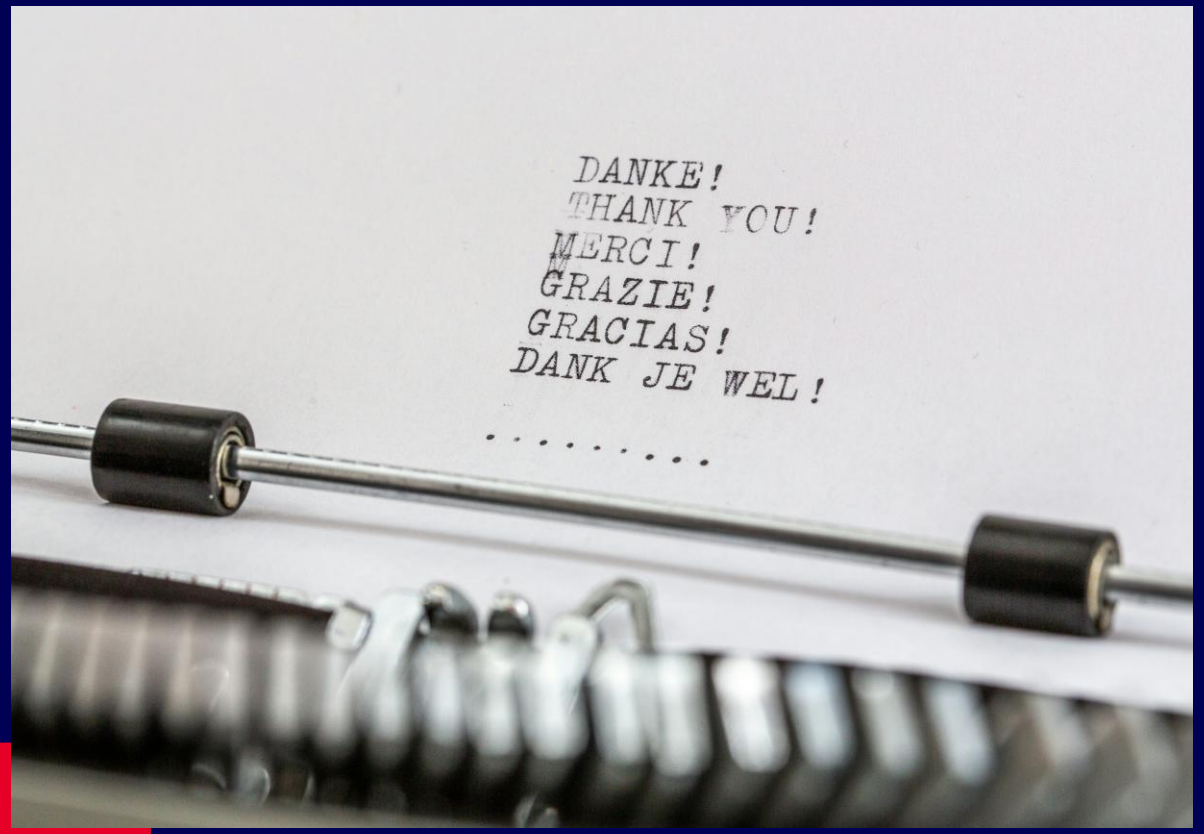
Thank You ☺



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